A SPIN–SCHLÄFLI OBSTRUCTION TO ±2 COSMETIC SURGERY

Sources: `notes/R8/hyp.md` and `notes/R8/hyp-code/`. K is a hyperbolic knot; by Ren we may assume this, and by Ravelomanana $Y_{\pm 2} = S^3_{\pm 2}(K)$ are then hyperbolic. $\mathrm{cs} \in \mathbb{R}/\mathbb{Z}$ denotes the real part of complex Chern–Simons in $SU(2)$ units. *Units (corrected; `notes/R8/examples.md`, `verify.md`).* SnapPy's `chern_simons` is $\pm 2\,\mathrm{cs}$ modulo $\frac{1}{2}$, so it determines cs only modulo $\frac{1}{4}$; $\mathrm{Im}\,\mathrm{cs} = -\mathrm{Vol}/4\pi^2$. The Riemannian Chern–Simons invariant is $2\,\mathrm{cs}$ modulo 1, i.e. cs modulo $\frac{1}{2}$. The first version of this note took cs = `chern_simons`; that was off by a factor 2.

1. Two exact facts

Lemma 1 (spin swap)**.** *$Y_{\pm 2}$ has two spin structures: s_0, which extends over the trace $X_{\pm 2}$, and s_1, which does not. If $\mathrm{Arf}(K) = 0$, then every orientation-preserving homeomorphism $Y_{-2} \to Y_2$ sends s_1 to s_0.*

Proof. By Kirby–Melvin, $\mu(Y_{\pm 2}, s_0) = \pm 1$ and $\mu(Y_{\pm 2}, s_1) = \mp 1 + 8\,\mathrm{Arf}(K)$ in $\mathbb{Z}/16$. A homeomorphism preserves μ, which forces the swap. □

Lemma 2 (half period)**.** *Let $\tilde{\rho}$ be an $SL_2(\mathbb{C})$ representation of $\pi_1(Y_{\pm 2})$, and let ε be the nontrivial character. Then $\mathrm{cs}(\tilde{\rho}\varepsilon) = \mathrm{cs}(\tilde{\rho}) + \frac{1}{2}$.*

Proof. Twisting by ε tensors the flat bundle with the flat line bundle L_ε. For Cheeger–Chern–Simons classes, $\hat{c}_2(E \otimes L) = \hat{c}_2(E) + \hat{c}_1(L) * \hat{c}_1(L)$. Here $\hat{c}_1(L_\varepsilon) * \hat{c}_1(L_\varepsilon)$ is the self-linking of the generator of $H_1 = \mathbb{Z}/2$, namely $\pm\frac{1}{2}$, equivalently $\langle \varepsilon^3, [Y] \rangle/2$ with $\varepsilon^3 \neq 0$. Checks: on $\mathbb{RP}^3$ the central representation has $\mathrm{cs} = \frac{1}{2}$; on $L(4,1)$, $\varepsilon^3 = 0$. □

Consequence. On a hyperbolic manifold, spin structures are exactly the SL_2 lifts of the holonomy, so $\mathrm{cs}(Y, s)$ is defined modulo 1. Lemma 2 gives $\mathrm{cs}(Y_{\pm 2}, s_1) = \mathrm{cs}(Y_{\pm 2}, s_0) + \frac{1}{2}$. A cosmetic homeomorphism is an isometry (Mostow) and carries s_1 to s_0 (Lemma 1). Hence

$$\mathrm{cs}(Y_2, s_0) - \mathrm{cs}(Y_{-2}, s_0) \equiv \tfrac{1}{2} \pmod 1.$$

The Riemannian (PSL_2, isometry) invariant fixes the left side modulo $\frac{1}{2}$, to 0. The spin structure supplies exactly the one missing bit, and it is the bit that a coincidence of PSL_2 data would leave open.

2. Computing the left side from the knot

Lemma 3. *Take the lift of the complete structure on E_K induced by the spin structure of S^3; it has $\mathrm{tr}\,\mu = \mathrm{tr}\,\lambda = -2$. Continue it along the cone deformation for slope $(N, 1)$ with N even. The trace of the filling slope is $-2\cos(\alpha/2)$, which reaches $+2$ at $\alpha = 2\pi$. So the lift descends to Y_N, and the spin structure it determines is s_0.*

Proposition 4 (complex Schläfli)**.** *Write $\tau_N(\alpha) = \mathrm{Im}\log\mathrm{hol}(\mu)$ along the cone path, with $\tau_N(0) = 0$. Then*

$$\mathrm{cs}(Y_N, s_0) \equiv c_E + \tfrac{N+3}{4} + \tfrac{1}{8\pi^2}\int_0^{2\pi} \tau_N\, d\alpha \pmod 1,$$

where c_E is the relative invariant of the complete S^3-lift. In particular $c_2 \equiv c_{-2}$. The coefficient is $\frac{1}{8\pi^2}$, not $\frac{1}{4\pi^2}$: the twist enters at half *the SnapPy rate.*

Proof and checks. Kirk–Klassen along the path, with $a = \frac{1}{2} + H_\mu/4\pi i$, $b = \frac{1}{2} + H_\lambda/4\pi i$, $NH_\mu + H_\lambda = i\alpha$; the endpoint terms in $h(2\pi)$ cancel against the solid-torus term. Changing the dual curve by m_{fill} changes the solid-torus term by an integer, so the earlier dual-curve worry dissolves. Independent mod-1 checks:

- exact $SU(2)$ analogue, 44 arcs on 12 torus knots: the difference is $\frac{1}{8\pi^2}\int(\tau_2 - \tau_{-2})$ exactly;
- Kirk–Klassen from an abelian character (exact base value 0) on the Riley curves of $6_2, 7_3, 6_3$: agreement to 10^{-6}, and $\mathrm{cs}(s_1) - \mathrm{cs}(s_0) = 0.5000000$, i.e. Lemma 2 numerically on a hyperbolic representation;
- closed amphichiral census: if $H_1 = \mathbb{Z}/2 \oplus \mathrm{odd}$, Rokhlin parity forces an orientation-reversing isometry to swap the spin structures, so $\mathrm{cs} \in \{\frac{1}{4}, \frac{3}{4}\}$. All 37 amphichiral census manifolds (seven of this type) have `chern_simons` $= 0$, which refutes cs = `chern_simons` and fits cs = ±`chern_simons`$/2$.

Theorem 5 (spin–Schläfli criterion)**.** *Let K be hyperbolic with $\mathrm{Arf}(K) = 0$. Suppose the cone paths from the complete structure to the ±2 fillings exist, and set*

$$\Delta_K = \frac{1}{4\pi^2}\int_0^{2\pi} \big(\tau_2 - \tau_{-2}\big)\, d\alpha.$$

If $S^3_2(K) \cong S^3_{-2}(K)$, then Δ_K is an odd integer*. (The Riemannian invariant alone gives $\Delta_K \in \mathbb{Z}$; the spin structure gives oddness.) Hence $|\Delta_K| < 1$ implies $S^3_2(K) \ncong S^3_{-2}(K)$. It suffices that $\sup|\tau_2| + \sup|\tau_{-2}| < 2\pi$.*

3. **Evidence and limits**

- *Survey of knots up to* 10 *crossings.* On the 217 knots whose cone paths are clean, $|\Delta| \le 0.135$ (median 0.014) and $\sup|\tau_2| + \sup|\tau_{-2}| \le 2.72 < 2\pi$. Over all knots surveyed, $\max|\Delta| = 0.385$ (10_{644}). The twist agent's "sharp negative" families reach $\Delta \approx \frac{1}{2} + \frac{1}{2(4k^2-1)} \le \frac{2}{3}$: no longer near the threshold.
- *Genus-*2 *knots with* $\Delta_K = 1$*, up to* 13 *crossings.* These are KT, $12n_{313}$, $12n_{430}$, $13n_{71}$, $13n_{1019}$, $13n_{1756}$, $13n_{1757}$ and $13n_{4591}$. For them $\Delta_K \in [0.028, 0.076]$. These checks are numerical and add nothing new, since Futer–Purcell–Schleimer already cover ≤ 19 crossings.
- *Heuristic.* At leading order in Neumann–Zagier, $\Delta_K \approx \frac{1}{2}\big[(2+x)/|\tau+2|^2 - (x-2)/|\tau-2|^2\big]$, where $\tau = x+iy$ is the cusp shape. This gives $|\Delta_K| \lesssim 1/(2\,\mathrm{Im}\,\tau)$, so a cosmetic pair would need a cusp with $\mathrm{Im}\,\tau \lesssim \frac{1}{2}$ or a large twist.
- *Known homeomorphic fillings* (`examples.md`). With the corrected units and genuine SL_2 lifts the formula reproduces every known identity: 788 amphichiral slopes (exact), Dunfield's $o9_{39009}$ pair (an instance of Ren's construction), a purely cosmetic family on the census manifold $o10_{150704}$, Ren's chiral Brunnian knot, $m135$, $t12051$. In every known pair the isolating twist difference is 0: the pair is induced by a symmetry of an ancestor, which pairs the lifts unmixed. A ± 2 pair in S^3 with $\mathrm{Arf} = 0$ would be the first *mixed* pairing, and needs a full odd period.
- *What it is.* It reduces the conjecture, for these knots, to an *a priori twist bound*: the cone paths must not accumulate a full (SnapPy) period of twist. The spin input is what is specific to S^3 and to ± 2. For $\pm 1/q$ there is one spin structure; for q odd the constants alone still force an odd twist difference, for q even there is no gap. Ren has shown that purely cosmetic fillings do exist on some multi-cusped hyperbolic manifolds, so any proof needs such input.
- *What it is not.* It is not uniform. The ± 2 slopes are short (normalised length about 2.8 for KT), outside the range where Hodgson–Kerckhoff guarantee that cone paths exist, and no a priori twist bound is known.
- *Gauge-theoretic reading.* The useful specialisation is $\theta = 0$: the real part of complex Chern–Simons, taken for the spin-lifted hyperbolic holonomy. Volume ($\theta = \pi/2$) cannot see orientation.